



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Development of a Productivity Management System for Remote Software Teams

Semester: Spring 25-26

<i>Group:</i>		<i>Section:</i>
SL	Student Name	Student ID
1.	MAISHA MAHJABIN	23-54978-3
2.	SADIA ISLAM ANUSHA	23-55670-3
3.	MD NAZMUL ISLAM	23-54433-3
4.	AL FARDIN SOWAD	23-54462-3

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1. PROJECT PROPOSAL

1.1 Background to the Problem

In recent years, remote software development has become increasingly common due to the availability of modern communication technologies and online collaboration tools. Many development teams now work from different locations instead of a single office. Although remote work provides flexibility and convenience, it also introduces several challenges such as difficulties in communication, task coordination, and monitoring project progress. Without a proper management system, it becomes difficult for team members to organize tasks, follow project schedules, and maintain effective collaboration. Team leaders or project managers may also face problems in assigning tasks, tracking productivity, and ensuring that project goals are achieved on time. These issues can reduce team efficiency and cause delays in software development. The primary target users of this system are remote software development teams, including student project teams, freelance developers, and small to medium-sized development groups who need a centralized platform to manage their projects and tasks. Secondary users include project managers, developers (team leaders), and administrators who are responsible for organizing work, monitoring sprint progress, and providing feedback to team members. To address these challenges, this project proposes a Productivity Management System for Remote Software Teams. The system will provide a centralized platform where users can create projects, manage product backlogs, plan sprints, assign tasks, track progress, and collaborate effectively. By integrating AI-assisted features and productivity monitoring tools, the system aims to improve coordination, efficiency, and overall productivity of remote software development teams.

1.2 Selection of Process Model

For this project, the Scrum model has been selected as the software development methodology. Scrum is an Agile approach that focuses on teamwork, flexibility, and continuous improvement during the development process. Instead of developing the whole system at once, Scrum divides the development process into small cycles called Sprints. Each Sprint produces a working part of the system. This allows the development team to review progress regularly and make improvements when necessary. Since the proposed system includes several modules such as project management, task tracking, sprint planning, and collaboration tools, Scrum is suitable because it allows the system to be developed step by step while receiving continuous feedback. The Scrum model is feasible for this project because it supports flexible and iterative development. The system requirements can be organized in a Product Backlog, which contains all the tasks and features needed for the system. During each Sprint, selected items from the backlog are developed and tested. This helps the team focus on smaller goals and complete tasks more efficiently. Scrum also encourages regular communication through meetings such as Sprint Planning, Daily Scrum, and Sprint Review, which helps the team stay coordinated and quickly solve problems. Another advantage of Scrum is that it allows changes to be added easily during development. If new features or improvements are required, they can be included in future Sprints without affecting the entire project. This makes Scrum a practical and effective model for this system.

2. Timeline Chart / Gantt Chart



3. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

3.1 Scopes and Features

3.1.1 User Authentication

Features:

- User Registration (Email/Password)
- Secure Login/Logout & Session Handling
- Profile Management (Name/Contact Updates)
- Role-Based Access Control

3.1.2 Project and Product Backlog Management

Features:

- Project Creation & Owner Assignment
- Centralized Product Backlog Management
- User Story Creation, Editing, and Deletion
- Backlog Prioritization

3.1.3 Sprint Planning

Features:

- Sprint Creation (Goals, Start/End Dates)
- Task Assignment within Sprints
- Effort Estimation (Hours/Remaining Work)
- AI-Assisted Sprint Scope Suggestions

3.1.4 Task Management

Features:

- Task Status Updates (To Do/In Progress/Done)
- Individual Task Boards for Developers
- Work Logging and Progress Notes
- Remaining Effort Tracking

3.1.5 Sprint Progress Monitoring

Features:

- Personal & Project Dashboards
- Real-time Completion Rate Tracking
- Sprint/Project Summary Reports
- Data Export (CSV/PDF)

3.1.6 Communication and Collaboration

Features:

- Threaded Task Comments
- In-app Notifications for @mentions
- Task Assignment & Status Change Alerts
- Activity Logs

3.1.7 Review and feedback

Features:

- Sprint Review Note Recording
- Stakeholder/Team Feedback Capture
- Searchable Historical Review Data

3.1.8 Help and support

Features:

- Searchable Help Center
- Categorized FAQ Section
- User Guidelines & Documentation
- Admin Content Management for Support

3.2 User Story Table

User (General):

As a/an	I want to	So that	Acceptance Criteria
User	register with email and password	I can create a secure account	Email must be valid and unique then successfully redirect to dashboard.
User	login and logout securely	I can protect my session	Correct credentials allow login. Wrong credentials show error. Logout blocks protected pages.
User	update my profile info	my team can identify me correctly	Name and contact cannot be empty. Changes must be saved and visible after refresh.
User	View my dashboard	I can track my tasks and notifications	User must be logged in. Assigned tasks and notifications must display correctly.
User	access help center	I can get guidance	Help content must be searchable. Categories must be visible.

Admin:

As a/an	I want to	So that	Acceptance Criteria
Admin	create projects and assign owners	development work can begin	Project name required. Owner must exist. Project appears in list after creation.
Admin	manage product backlog	requirements are organized	Can add/edit/delete stories. Priority must be selectable. Changes saved instantly.
Admin	maintain role-based access	users access correct features	Unauthorized access shows error (404). Roles restrict features properly.
Admin	manage help content	documentation stays updated	Admin can add/edit/delete articles. Changes visible to users immediately.
Admin	export reports	I can review progress	Reports must download in CSV/PDF. Data must match dashboard.

Developer:

As a/an	I want to	So that	Acceptance Criteria
Developer	view assigned tasks	I know my work	All assigned tasks appear in board.
Developer	update task status	team knows my progress	Status must be To Do/In Progress/Done. Changes reflect instantly.
Developer	log work and remaining hours	progress is tracked	Hours must be positive. Updates reflect in sprint charts.
Developer	comment on tasks	I can collaborate	Comments must save instantly. User can edit/delete own comments.
Developer	receive notifications	I stay updated	Notifications appear for assignment and mentions in real time.

Project Manager (PM):

As a/an	I want to	So that	Acceptance Criteria
PM	create sprints	work is organized	Start and end date required. Sprint saved with status “Planned”.
PM	assign tasks	work is distributed	Task must have assignee. Task appears in sprint board.
PM	prioritize backlog	important work done first	Priority levels must be selectable. Sorted correctly.
PM	monitor progress	I track performance	Dashboard shows completion rate and status distribution.
PM	record feedback	improve future sprints	Notes must be saved and linked to sprint.
PM	use AI suggestions	optimize planning	AI suggestions must show with explanation. PM can accept/override.

AI assistant:

As a/an	I want to	So that	Acceptance Criteria
System (AI)	recommend assignees	workload is balanced	Suggestions based on workload and skill. PM can override.
System (AI)	suggest sprint scope	planning improves	Suggested tasks based on capacity.
System (AI)	detect risks	delays are avoided	At-risk tasks flagged with reason. Alerts shown on dashboard.

3.3 Requirements Traceability Matrix

3.3.1 Functional Requirements

3.3.1.1 USER

Req ID	Requirement Description	Module / Scope
FR-01	Allow the user to register with email and password.	Submitting valid details creates an account and confirms success.
FR-02	Allow secure login/logout	Correct credentials start a session; after logout, protected pages are blocked.
FR-03	Allow the user to update profile (name, contact).	Saved changes appear on profile and persist after refresh.
FR-04	Show a personal dashboard with assigned tasks and notifications.	Dashboard lists current tasks and latest notifications.
FR-05	Provide access to Help Center and FAQ.	Help is searchable and displays categorized guides.

3.3.1.2 ADMIN

Req ID	Requirement Description	Module / Scope
FR-06	Create projects and set owners.	New project appears on project list with owner set.
FR-07	Manage product backlog at a system/project level.	Can add/edit/priority user stories
FR-08	Maintain role-based access control	Restricted features return 403 to unauthorized roles.

FR-09	Provide Help & Support content (guides, FAQ).	Admin can publish and update help articles.
FR-10	Access reports (sprint/project summaries) and export.	Export to CSV/PDF with timestamp.

3.3.1.3 DEVELOPER

Req ID	Requirement Description	Module / Scope
FR-11	View assigned tasks	The developer sees all assigned tasks on their board.
FR-12	Update task status (To Do / In Progress / Done).	Status change persists and appears across dashboards.
FR-13	Log work / update progress (remaining hours, notes).	Remaining effort updates reflect sprint charts.
FR-14	Comment on tasks for collaboration and feedback.	Post/edit (short window) and delete own comments visible in thread.
FR-15	Receive notifications (assignment, @mentions, status).	Triggering events show in an in-app notifications panel.

3.3.1.4 PROJECT MANAGER (PM)

Req ID	Requirement Description	Module / Scope
FR-16	Create sprints with start/end dates and goals.	Sprint saved under project with status "Planned."

FR-17	Create and assign tasks within sprints.	Tasks appear in “To Do” with assigned owner and estimate.
FR-18	Manage backlog (add/edit/prioritize).	Backlog sorted by priority; changes persist.
FR-19	Monitor sprint progress via dashboards and reports.	Dashboards show completion rate, status distribution, and throughput.
FR-20	Record sprint review notes and capture feedback.	Notes/feedback attach to sprint or tasks and are reportable.
FR-21	Use AI suggestions for sprint scope and task assignment.	“AI Suggest” proposes scope/assignees with rationale; PM can override.

3.3.1.5 AI ASSISTANT

Req ID	Requirement Description	Module / Scope
FR-22	Recommend assignees based on workload/skill.	Shows proposed assignees and rationale; PM can accept/override.
FR-23	Suggest sprint scope based on capacity.	Proposes backlog items with predicted completion likelihood.
FR-24	Predict to schedule risks and flag at-risk tasks.	Flags appear on dashboard with reason codes (e.g., over-capacity).

3.3.2 Non-Functional Requirements

3.3.2.1 USER

Req ID	Requirement Description	Module / Scope
NFR-01	Strong auth & session handling.	Security
NFR-02	First-time users can onboard quickly.	Usability
NFR-03	Inclusive UI.	Accessibility
NFR-04	Work across time zones.	Localization

3.3.2.2 ADMIN

Req ID	Requirement Description	Module / Scope
NFR-05	Enforce HTTPS/TLS and RBAC.	Security
NFR-06	Keep activity logs.	Auditability
NFR-07	Docs & test coverage.	Maintainability

3.3.2.3 DEVELOPER

Req ID	Requirement Description	Module / Scope
NFR-08	Snappy task updates.	Performance
NFR-09	Data integrity updates.	Reliability

NFR-10	Efficient board interactions.	Usability
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3.3.2.4 PROJECT MANAGER

Req ID	Requirement Description	Module / Scope
NFR-11	Dashboard freshness.	Performance
NFR-12	Team and backlog scale.	Scalability
NFR-13	Monitoring & alerts.	Observability
NFR-14	Retention policies.	Privacy

3.3.2.5 AI ASSISTANT

Req ID	Requirement Description	Module / Scope
NFR-15	Explainable suggestions.	Transparency
NFR-16	Human-in-the-loop.	Safety & Ethics
NFR-17	Reasonable inference time.	Performance

4. SOFTWARE DESIGN

4.1 Use Case Diagram

The use case diagram for the Productivity Management System for Remote Software Teams illustrates the functional interactions between four primary actors: the General User, Admin, Developer, and Project Manager, alongside an integrated AI Assistant. The General User handles foundational account actions such as signing up, signing in, and managing their personal profile, while also having access to help and support modules. The Admin oversees high-level system operations, including project creation, maintaining role-based access control, and managing the overall product backlog to ensure requirements are met. On the execution side, the Developer interacts with the system to view assigned tasks, update their status, log work progress, and collaborate with teammates through threaded comments. The Project Manager acts as the primary orchestrator, responsible for sprint planning, task assignment, and monitoring real-time progress through dashboards to identify throughput and completion rates. Finally, the AI Assistant serves as a specialized support actor that provides data-driven suggestions for sprint scopes, recommends task assignees based on skill levels, and predicts potential schedule risks to help the team avoid delays.

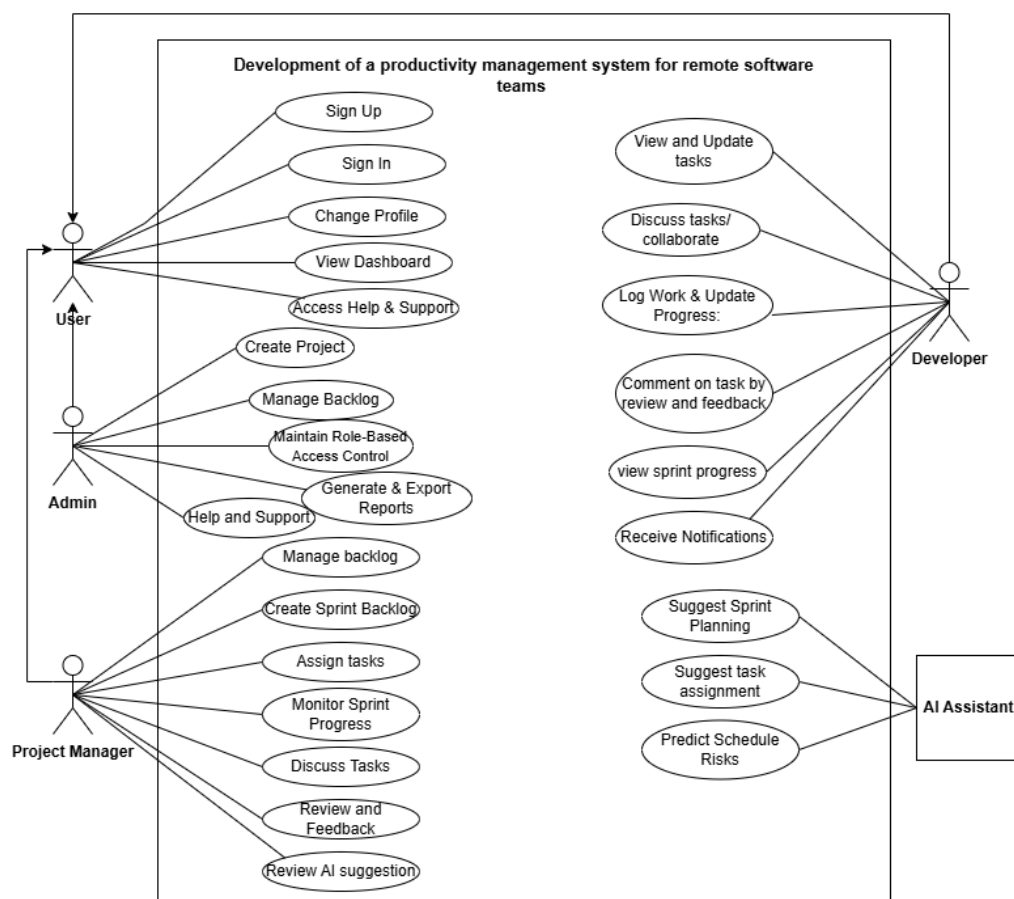


Fig: Use Case Diagram for the system

Class Diagram

The class diagram for the Productivity Management System represents the structural design of the system by showing the main classes and their relationships. The central User class is extended by Admin, Developer, and Project Manager, each performing specific roles. The Project class contains the Backlog and multiple Sprints, where backlog items are managed and sprints are planned with defined goals and timelines. Each sprint consists of multiple Tasks, which are assigned to developers and updated based on progress. The Project Manager oversees task assignment and sprint planning, while developers execute tasks and update their status. Additionally, an AI Assistant supports the system by providing task recommendations and risk predictions. Overall, the diagram illustrates how different components interact to ensure efficient project and task management.

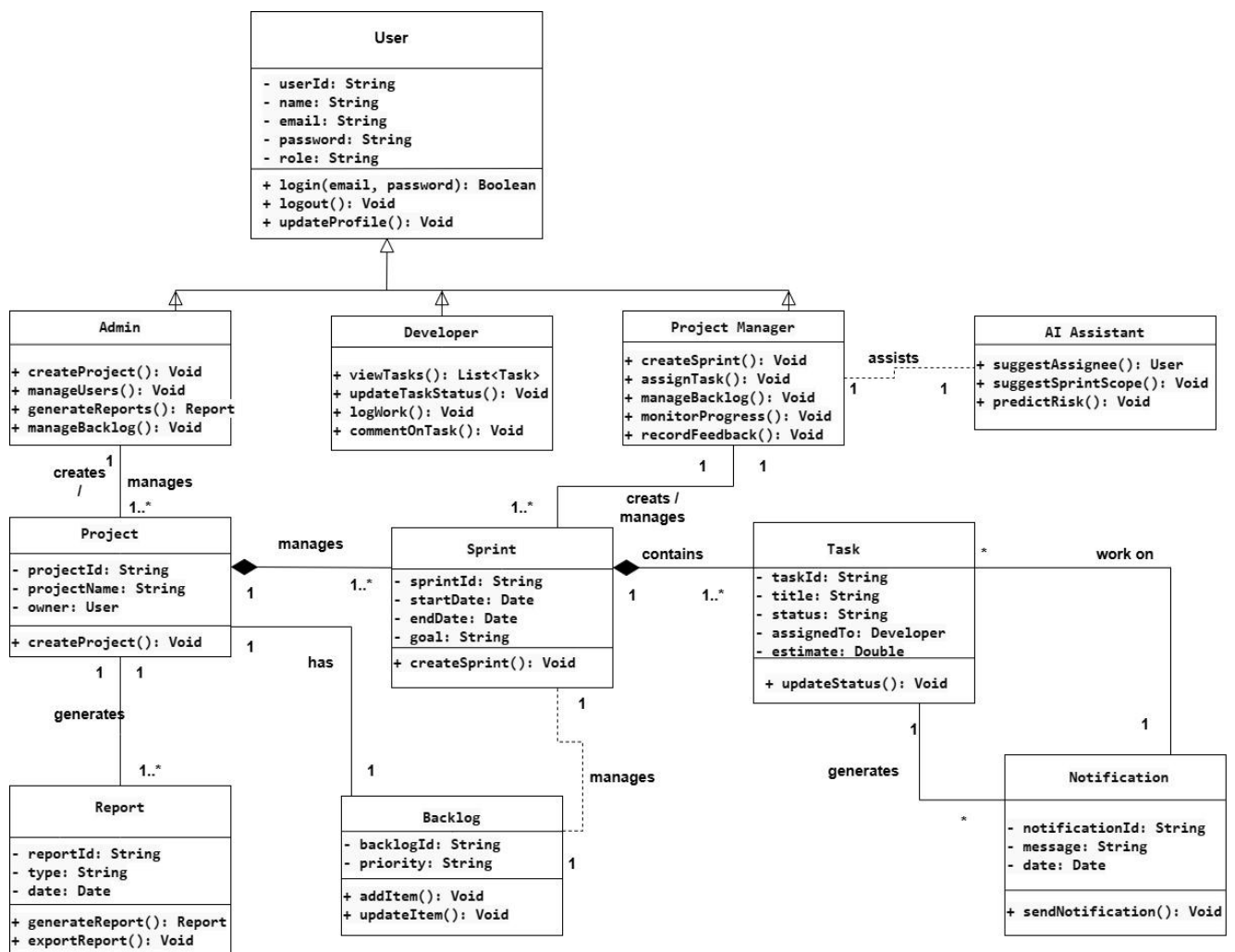


Fig: Class Diagram for the system

4.2 UI / Wireframe Design using Figma

Wireframe Designs:

Maisha Mahjabin (23-54978-3)

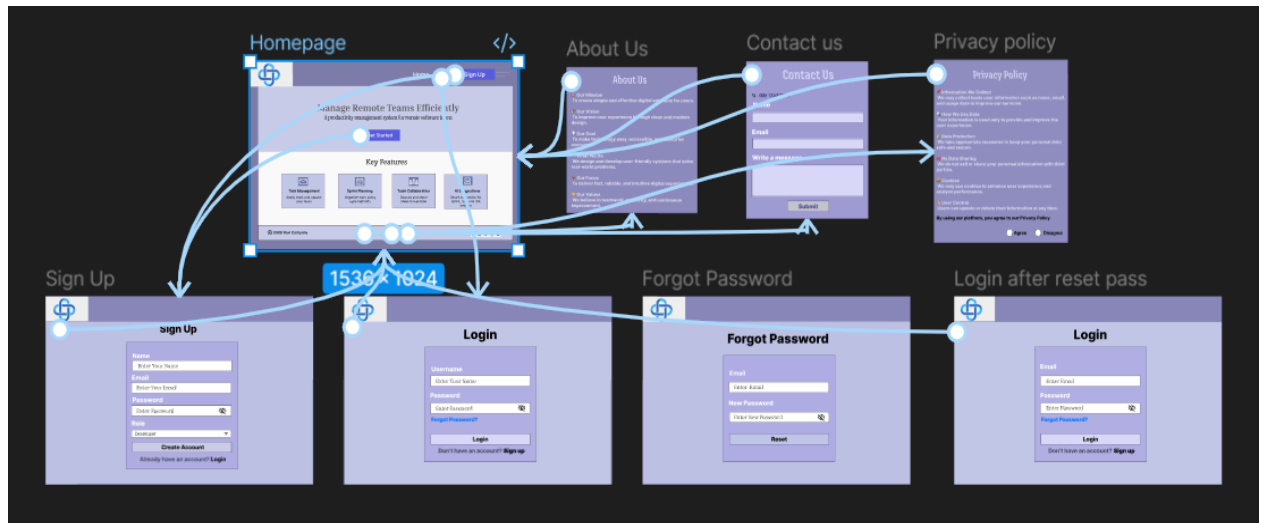


Fig: Homepage wiring



Fig: Authentication pages wiring

Sadia Islam Anusha (23-55670-3)

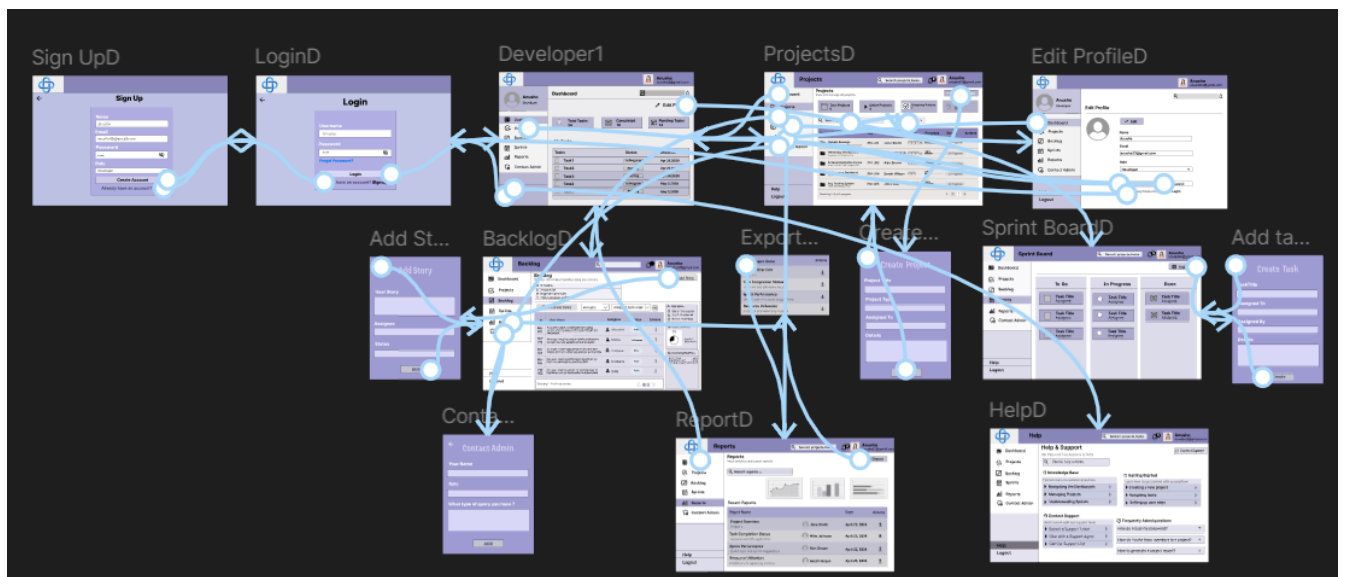


Fig: Developer Dashboard wiring

MD Nazmul Islam (23-54433-3)

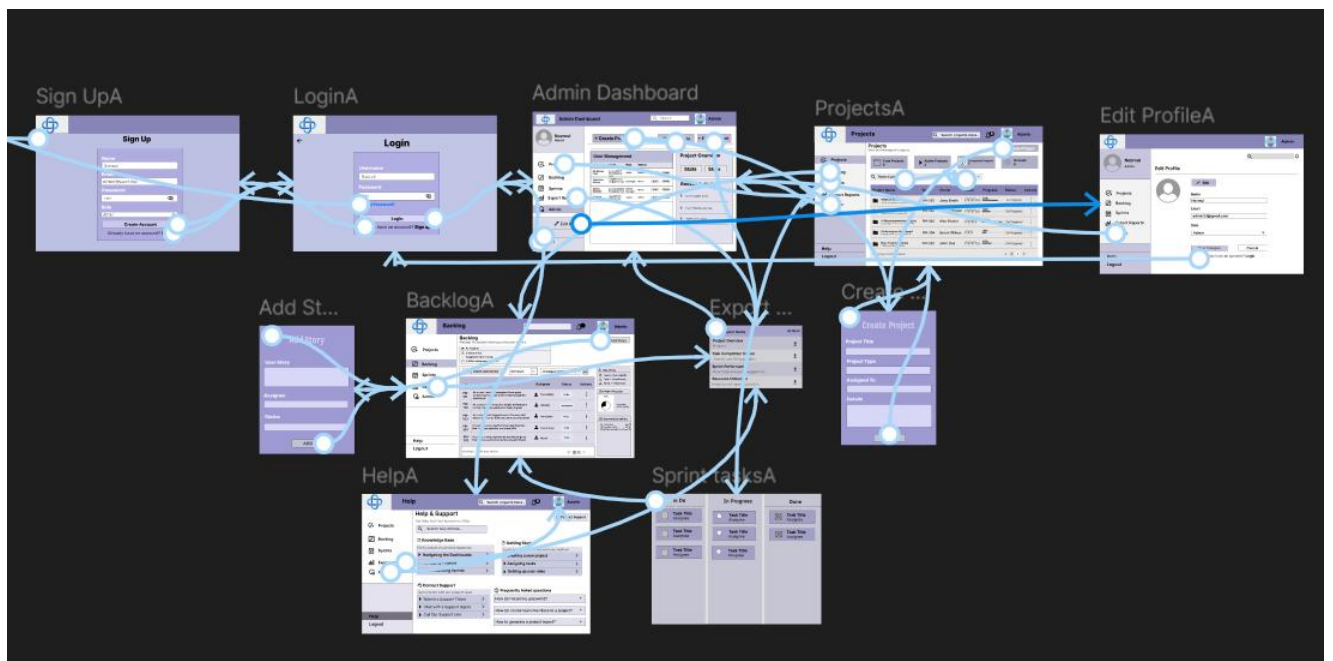


Fig: Admin Dashboard wiring

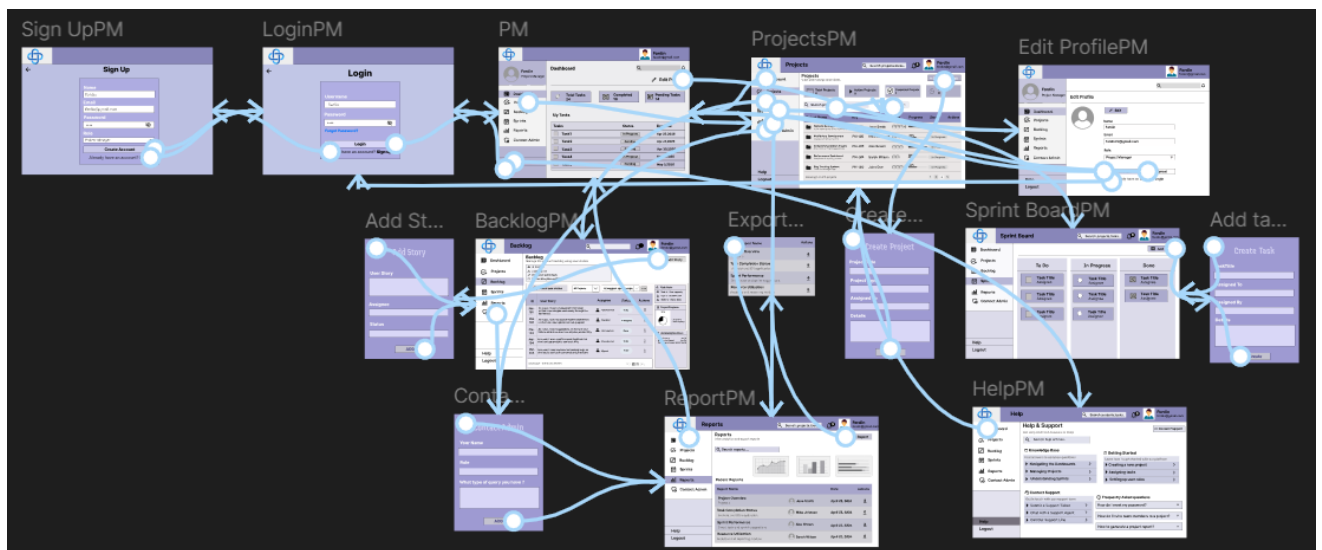


Fig: Project Manager Dashboard wiring

UI Designs:

Figma Link:

<https://www.figma.com/design/AjkdraQwXfVF4tj5kWmOP5/Design?node-id=0-1&t=ayWz3Jxqmardd5iX-1>

By Maisha Mahjabin (23-54978-3)

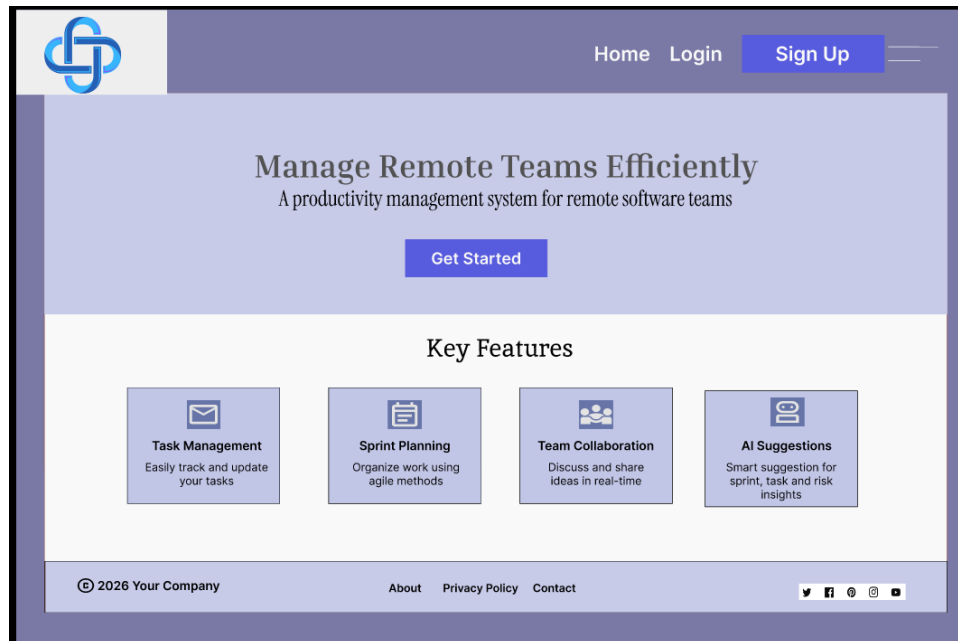


Fig: Homepage

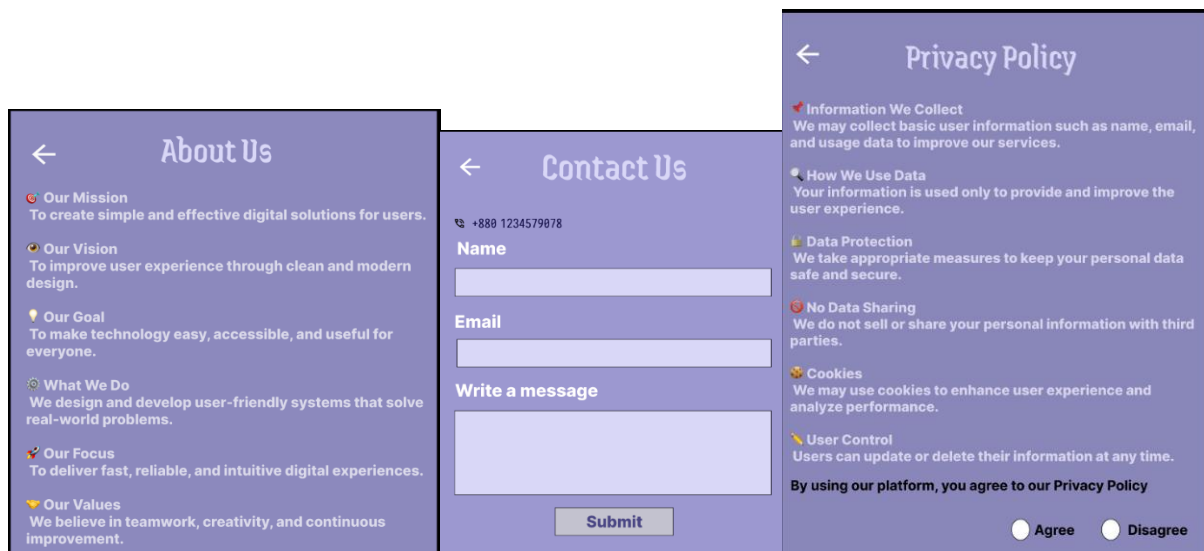
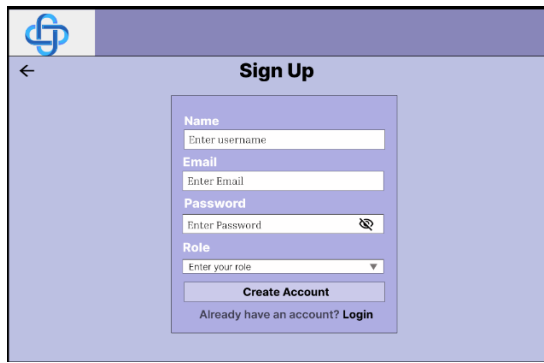


Fig: About us, Contact us and Privacy policy page



Sign Up

Name
Enter username

Email
Enter Email

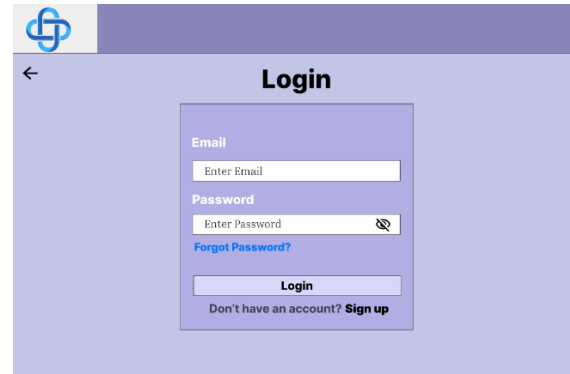
Password
Enter Password

Role
Enter your role

Create Account

Already have an account? [Login](#)

Fig: Signup page



Login

Email
Enter Email

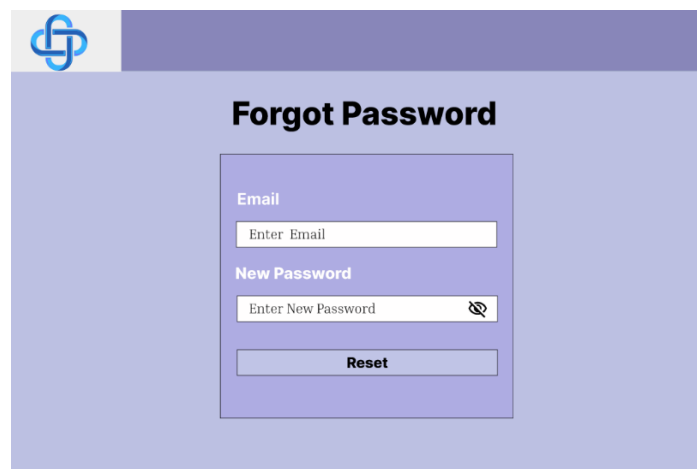
Password
Enter Password

[Forgot Password?](#)

Login

Don't have an account? [Sign up](#)

Fig: Login page



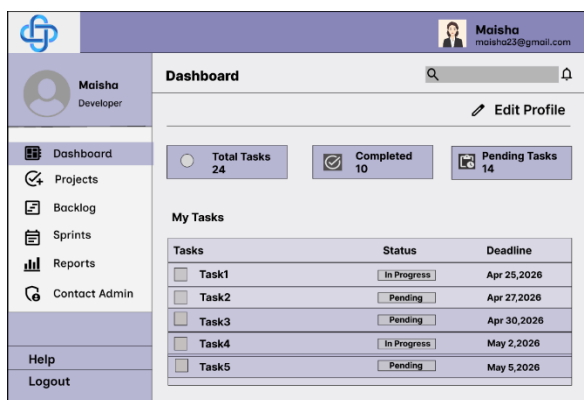
Forgot Password

Email
Enter Email

New Password
Enter New Password

Reset

Fig: Forget Password page



Maisha Developer

Dashboard

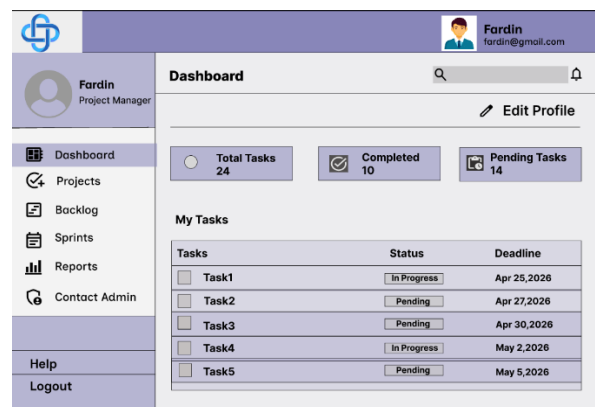
Total Tasks 24

Completed 10

Pending Tasks 14

My Tasks

Tasks	Status	Deadline
Task1	In Progress	Apr 25, 2026
Task2	Pending	Apr 27, 2026
Task3	Pending	Apr 30, 2026
Task4	In Progress	May 2, 2026
Task5	Pending	May 5, 2026



Fordin Project Manager

Dashboard

Total Tasks 24

Completed 10

Pending Tasks 14

My Tasks

Tasks	Status	Deadline
Task1	In Progress	Apr 25, 2026
Task2	Pending	Apr 27, 2026
Task3	Pending	Apr 30, 2026
Task4	In Progress	May 2, 2026
Task5	Pending	May 5, 2026

Fig: Developer, Project manager Dashboard

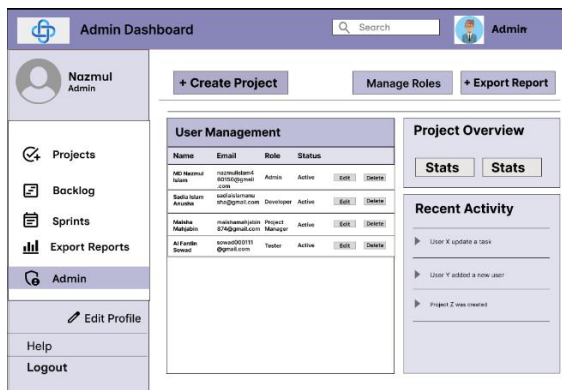


Fig: Admin Dashboard

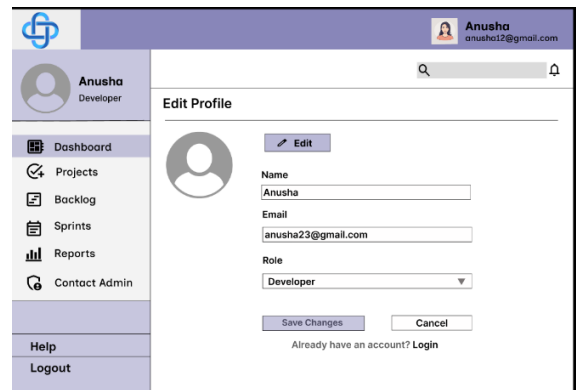


Fig: Edit Profile Page

By AL Fardin Sowad (23-54462-3)

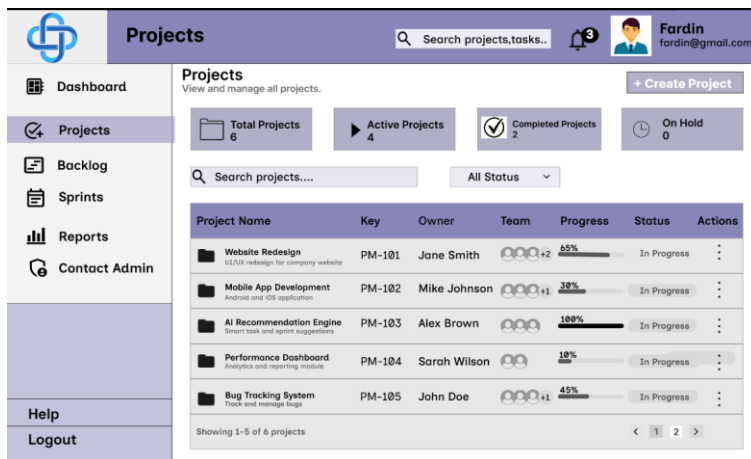


Fig: Projects panel

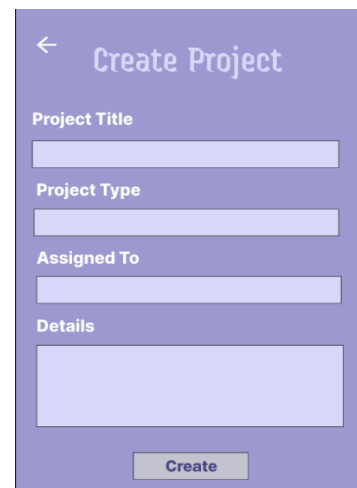


Fig: Create Project form

By Md Nazmul Islam (23-54433-3)

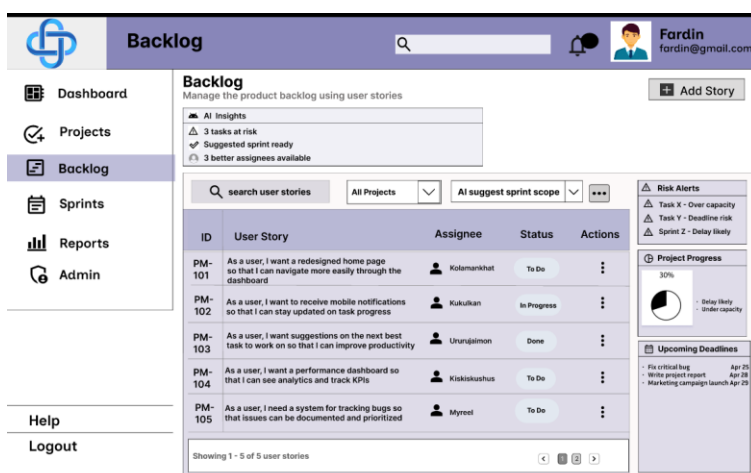


Fig: Backlog Panel

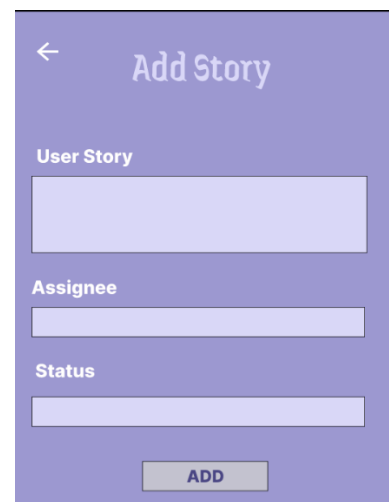


Fig: Add Story Form

By Sadia Islam Anusha (23-55670-3)

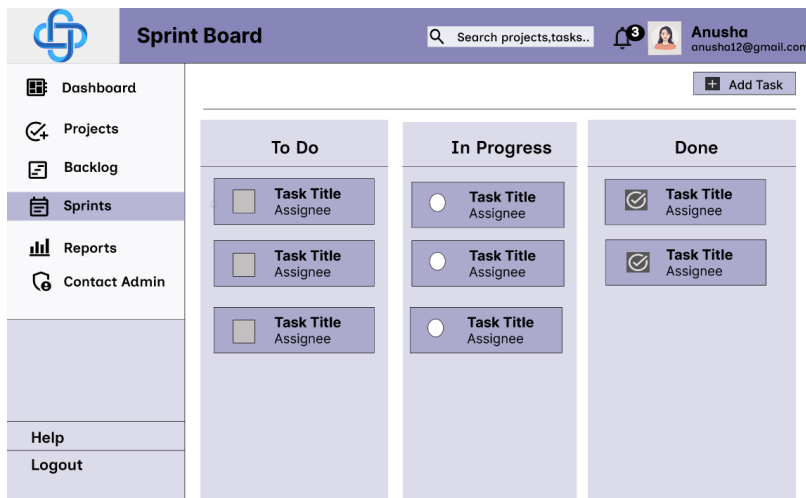


Fig: Sprints Panel

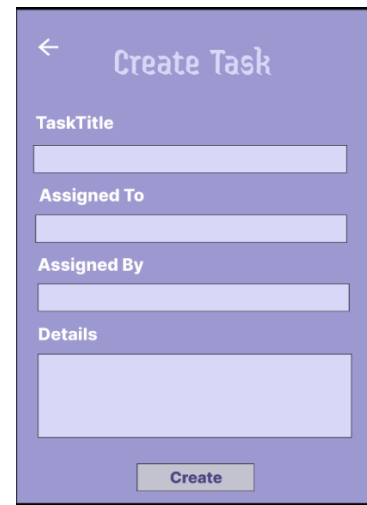


Fig: Add Task

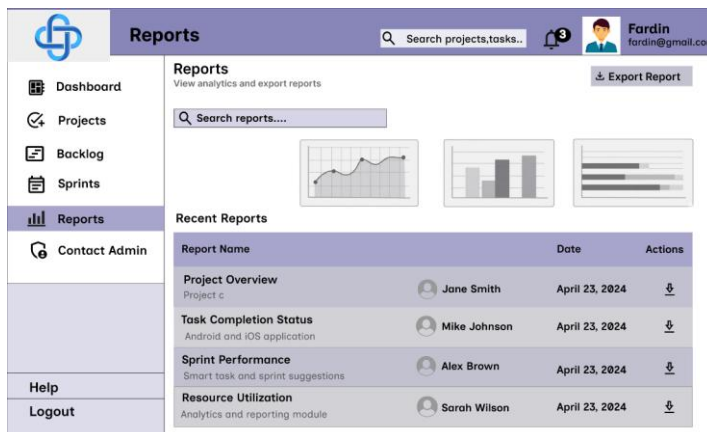


Fig: Reports Panel

Report Name	Actions
Project Overview Project c	⬇
Task Completion Status Android and iOS application	⬇
Sprint Performance Smart task and sprint suggestions	⬇
Resource Utilization Analytics and reporting module	⬇

Fig: Export Reports Page

By Nazmul Islam (23-54433-3)

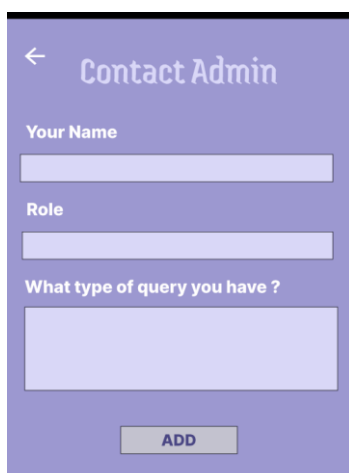


Fig: Contact Admin Form

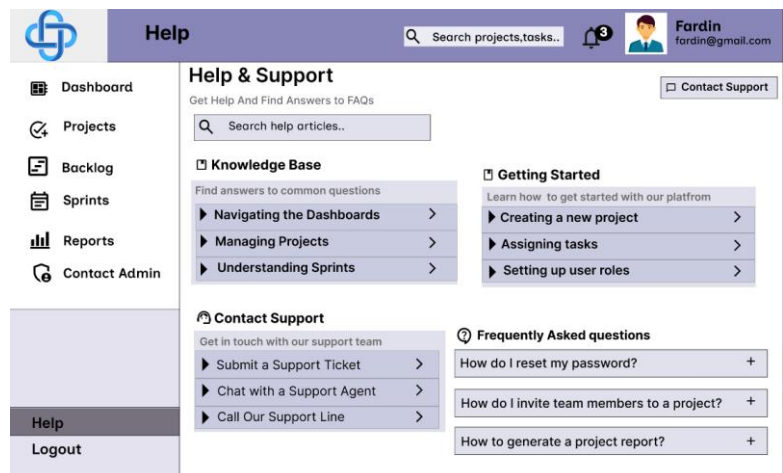


Fig: Help and support panel

5. GIT WORKFLOW

Repository created, initialization local to remote and named the default branches and implemented a feature

By Maisha Mahjabin (23-54978-3)

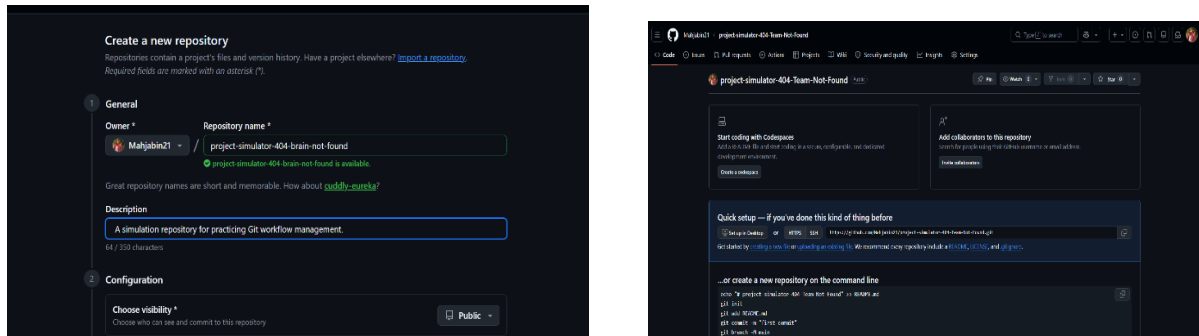


Fig: Created a repository

```
Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project (master)
$ mkdir project-simulator

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project (master)
$ cd project-simulator

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (master)
$ git init
Initialized empty Git repository in C:/Users/User/Downloads/project/project-simulator/.git/

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (master)
$ git init
Initialized empty Git repository in C:/Users/User/Downloads/project/project-simulator/.git/

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (master)
$ echo "# Project Simulator" > README.md

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (master)
$ echo "# Changelog" > CHANGELOG.md

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (master)
$ echo "# Implemented Features" > PROJECT_FEATURES.md

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (master)
$ git add README.md CHANGELOG.md PROJECT_FEATURES.md
warning: in the working copy of 'CHANGELOG.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'PROJECT_FEATURES.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
```



```

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simu
aster)
$ git commit -m "chore: initial project setup and documentation"
[master (root-commit) 79edf93] chore: initial project setup and documenta
3 files changed, 3 insertions(+)
create mode 100644 CHANGELOG.md
create mode 100644 PROJECT_FEATURES.md
create mode 100644 README.md

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
aster)
$ git branch -M main

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git branch dev

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git branch stage

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git branch
  dev
* main
  stage

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git remote add origin https://github.com/Mahjabin21/-project-simulator-404-bra
in-not-found.git

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 388 bytes | 194.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Mahjabin21/-project-simulator-404-brain-not-found.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git push -u origin dev
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/Mahjabin21/-project-simulator-404-brain-not-foun
d/pull/new/dev
remote:
To https://github.com/Mahjabin21/-project-simulator-404-brain-not-found.git
 * [new branch]      dev -> dev
branch 'dev' set up to track 'origin/dev'.

Maisha Mahjabin@DESKTOP-HSVFOCH MINGW64 ~/Downloads/project/project-simulator (m
ain)
$ git push -u origin stage
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'stage' on GitHub by visiting:
remote:   https://github.com/Mahjabin21/-project-simulator-404-brain-not-foun
d/pull/new/stage
remote:
To https://github.com/Mahjabin21/-project-simulator-404-brain-not-found.git
 * [new branch]      stage -> stage
branch 'stage' set up to track 'origin/stage'.

```

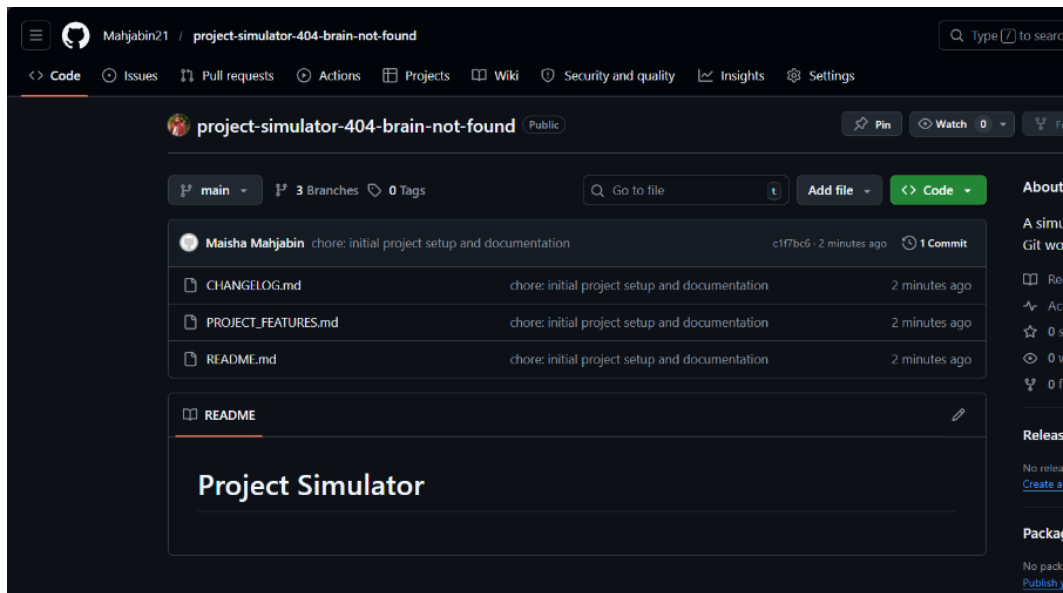



Fig: Initialized the Repository Locally and Connected the Local Repository to the Remote

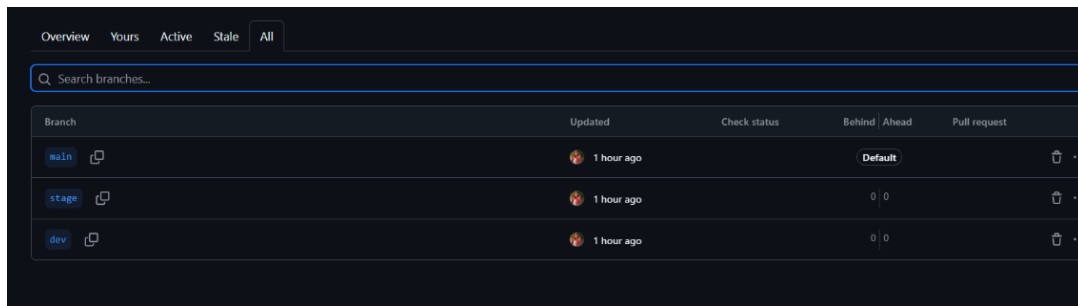


Fig: Renamed the Default Branches

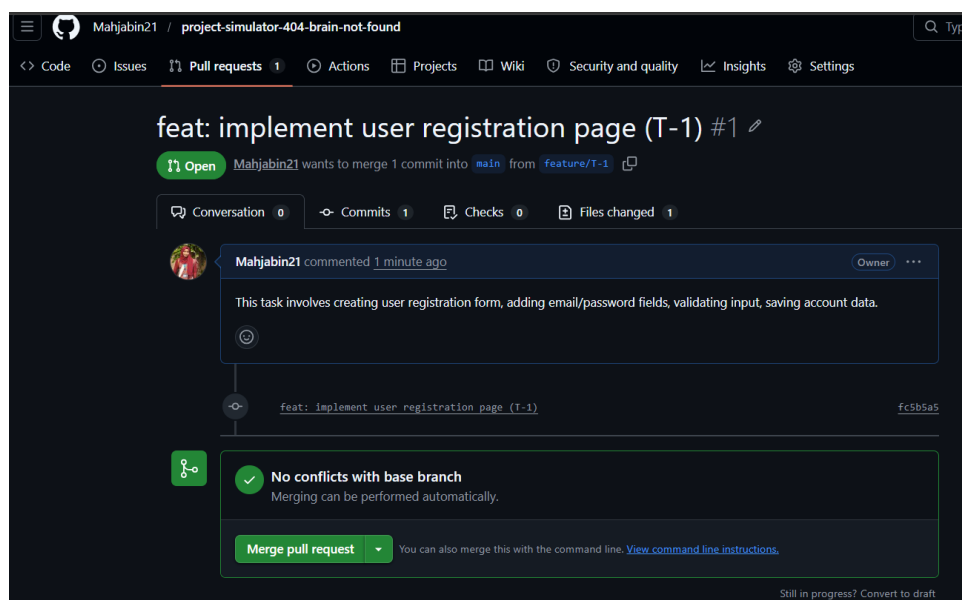


Fig: Push and pull request for implementation of **Task1 User Registration Page**

After cloning and pull requested by other group members:

By Sadia Islam Anusha (23-55670-3)

```
$ git clone https://github.com/Mahjabin21/project-simulator-404-brain-not-found
git
Cloning into 'project-simulator-404-brain-not-found'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (22/22), done.
remote: Compressing objects: 100% (19/19), done.
remote: Total 22 (delta 5), reused 7 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (22/22), 6.72 KiB | 1.12 MiB/s, done.
Resolving deltas: 100% (5/5), done.
```

```
$ git config user.name "Sadia Islam Anusha"
git config user.email "sadiaislamanusha@gmail.com"
```

```
$ git add PROJECT_FEATURES.md
git commit -m "feat: implement user login page (T-2)"
[feature/T-2 (root-commit) 717308f] feat: implement user login page (T-2)
1 file changed, 5 insertions(+)
create mode 100644 Downloads/soft/PROJECT_FEATURES.md
```

```
$ git push -u origin feature/T-2
```

The screenshot shows a GitHub pull request interface. At the top, the title is "feat: implement user login page (T-2) #2". Below the title, it says "Mahjabin21 wants to merge 1 commit into dev from feature/T-2". There are tabs for "Conversation" (0), "Commits" (1), "Checks" (0), and "Files changed" (1). A comment from Mahjabin21, posted 19 hours ago, states: "This task involves creating a login form component with email and password fields, including validation and error handling." Below the comment, there is a commit preview for "feat: implement user login page (T-2)". At the bottom, a green box indicates "No conflicts with base branch" and "Merging can be performed automatically."

Fig: Push and pull request for implementation of Task2 Login Page

By AL FARDIN SOWAD (23-54462-3)

```
$ git clone https://github.com/Mahjabin21/project-simulator-404-brain-not-found.git
Cloning into 'project-simulator-404-brain-not-found'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 8 (delta 0), reused 8 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), done.
```

```
$ git config user.name "fardin-sowad"
git config user.email "sowad000111@gmail.com"
```

```
$ git checkout main
git pull origin main
git checkout dev
git pull origin dev
git checkout stage
git pull origin stage
Already on 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch          main          -> FETCH_HEAD
Already up to date.
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch          dev           -> FETCH_HEAD
Already up to date.
branch 'stage' set up to track 'origin/stage'.
Switched to a new branch 'stage'
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch          stage         -> FETCH_HEAD
Already up to date.
```

```
-brain-not-found (stage)
$ git checkout dev
git pull origin dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch          dev           -> FETCH_HEAD
Already up to date.
```

```
$ git checkout -b feature/T-3
Switched to a new branch 'feature/T-3'
```

```
$ echo "## T-3: Implement Profile Edit Page" >> PROJECT_FEATURES.md
```

```
$ git checkout -b feature/T-3
Switched to a new branch 'feature/T-3'
```

```
-brain-not-found (feature/T-3)
$ echo "## T-3: Implement Profile Edit Page" >> PROJECT_FEATURES.md
echo "This task manages profiles by adding editable fields, updating name/contact and saving changes." >> PROJECT_FEATURES.md
echo "***Status: Implemented**" >> PROJECT_FEATURES.md
echo "" >> PROJECT_FEATURES.md
```

```

-brain-not-found (feature/T-3)
$ git add PROJECT_FEATURES.md
git commit -m "feat: implement edit profile page (T-3)"
warning: in the working copy of 'PROJECT_FEATURES.md', LF will be replaced by CR
LF the next time Git touches it
[feature/T-3 05c6597] feat: implement edit profile page (T-3)
1 file changed, 4 insertions(+)

-brain-not-found (feature/T-3)
$ git push -u origin feature/T-3
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 547 bytes | 547.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/T-3' on GitHub by visiting:
remote:   https://github.com/Mahjabin21/project-simulator-404-brain-not-found
/pull/new/feature/T-3
remote:
To https://github.com/Mahjabin21/project-simulator-404-brain-not-found.git
 * [new branch]      feature/T-3 -> feature/T-3
branch 'feature/T-3' set up to track 'origin/feature/T-3'.

```

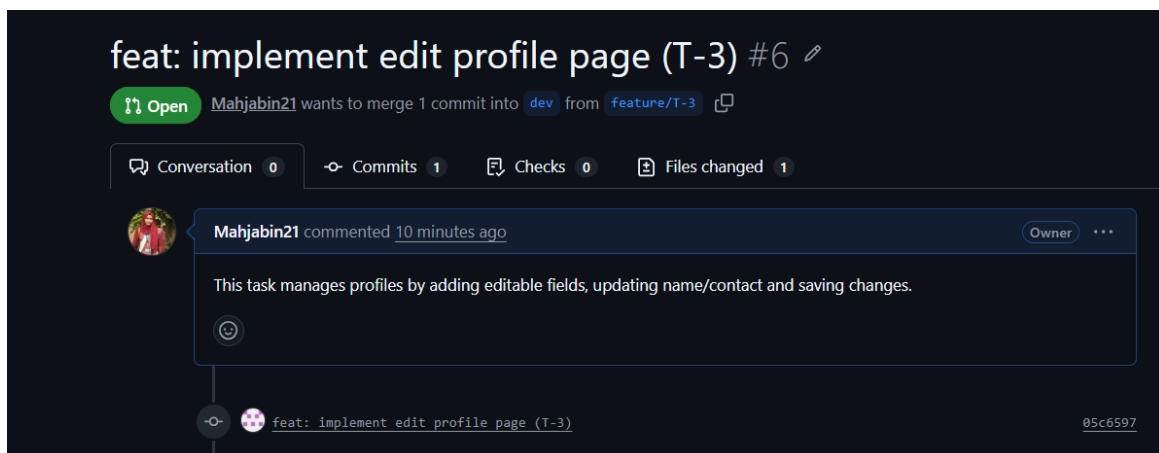


Fig: Push and pull request for implementation of **Task3 Edit Profile Page**

By Md Nazmul Islam (23-54433-3)

```

$ git clone https://github.com/Mahjabin21/project-simulator-404-brain-not-found.git
Cloning into 'project-simulator-404-brain-not-found'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 8 (delta 0), reused 8 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), done.

$ git config user.name "MD Nazmul Islam"
git config user.email "nazmulislam460150@gmail.com"

```



```

-brain-not-found (main)
$ git checkout main
git pull origin main
git checkout dev
git pull origin dev
git checkout stage
git pull origin stage
Already on 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch      main      -> FETCH_HEAD
Already up to date.
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch      dev       -> FETCH_HEAD
Already up to date.
branch 'stage' set up to track 'origin/stage'.
Switched to a new branch 'stage'
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch      stage     -> FETCH_HEAD
Already up to date.

```

```

-brain-not-found (stage)
$ git checkout dev
git pull origin dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.
From https://github.com/Mahjabin21/project-simulator-404-brain-not-found
 * branch      dev       -> FETCH_HEAD
Already up to date.

```

```

-brain-not-found (dev)
$ git checkout -b feature/T-5
Switched to a new branch 'feature/T-5'

```

```

-brain-not-found (feature/T-5)
$ echo "## T-5: Implement Project Creation form" >> PROJECT_FEATURES.md

```

```

-brain-not-found (dev)
$ git checkout -b feature/T-5
Switched to a new branch 'feature/T-5'

```

```

-brain-not-found (feature/T-5)
$ echo "## T-5: Implement Project Creation form" >> PROJECT_FEATURES.md
echo "This task creates project form, assigns project owner and saves project" >
> PROJECT_FEATURES.md
echo "***Status: Implemented***" >> PROJECT_FEATURES.md
echo "" >> PROJECT_FEATURES.md

```

```

-brain-not-found (feature/T-5)
$ git add PROJECT_FEATURES.md
git commit -m "feat: implement project creation form (T-5)"
warning: in the working copy of 'PROJECT_FEATURES.md', LF will be replaced by CR
LF the next time Git touches it
[feature/T-5 6872bb6] feat: implement project creation form (T-5)
1 file changed, 4 insertions(+)

```

```

$ git push -u origin feature/T-5
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 491 bytes | 491.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/T-5' on GitHub by visiting:
remote:   https://github.com/Mahjabin21/project-simulator-404-brain-not-found
/pull/new/feature/T-5
remote:
To https://github.com/Mahjabin21/project-simulator-404-brain-not-found.git
 * [new branch]      feature/T-5 -> feature/T-5
branch 'feature/T-5' set up to track 'origin/feature/T-5'.

```



Fig: Push and pull request for implementation of **Task5 Project Creation Page**

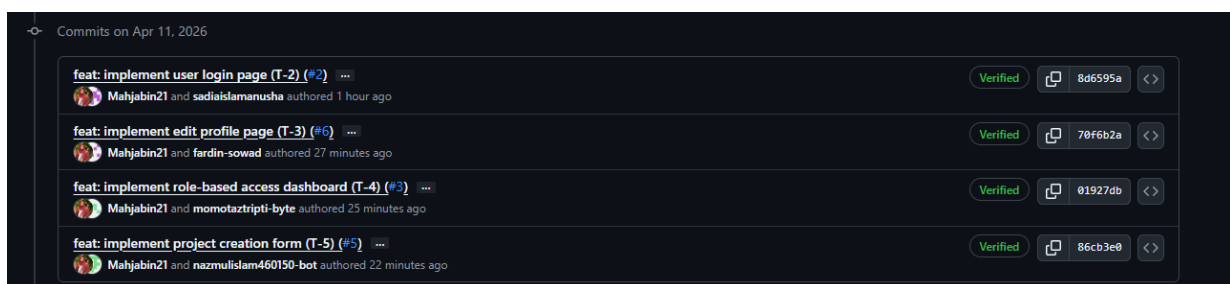


Fig: All other members commit

Review, Squash and merge the features

By Maisha Mahjabin (23-54978-3)

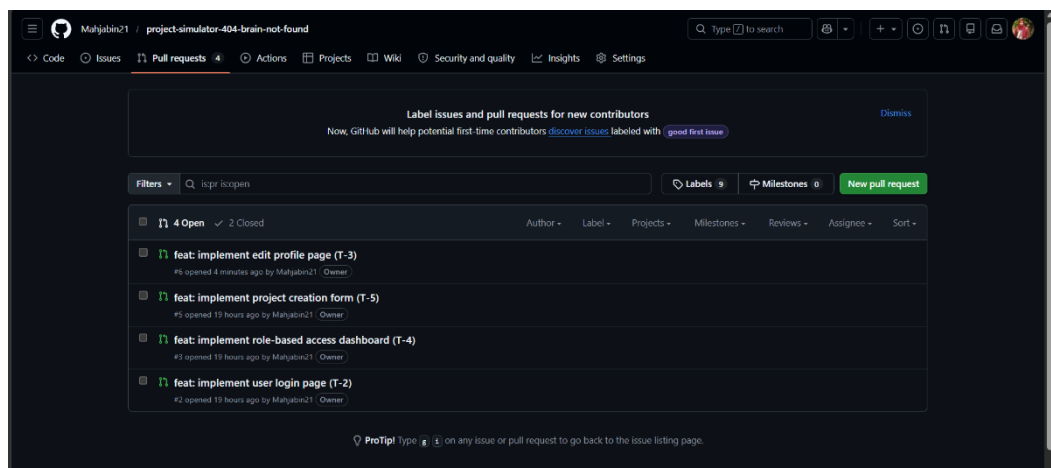


Fig: After getting the pull requests by other members

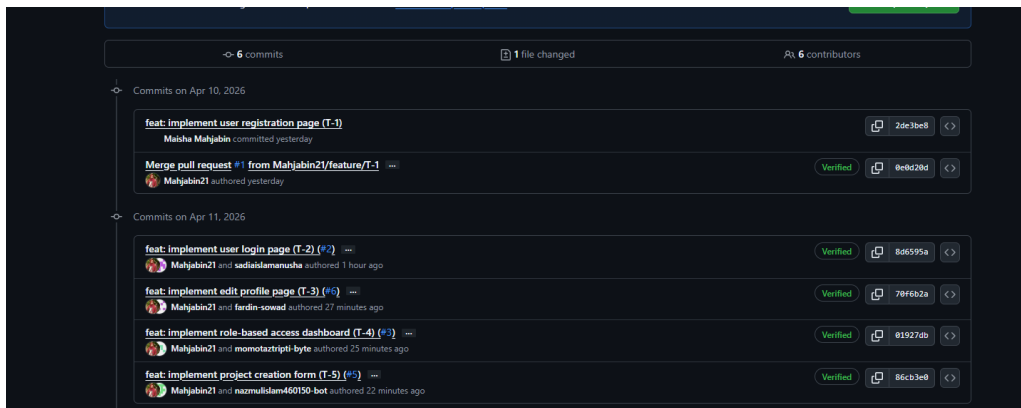


Fig: Number of commits after squashing and merging the pull requests by other members

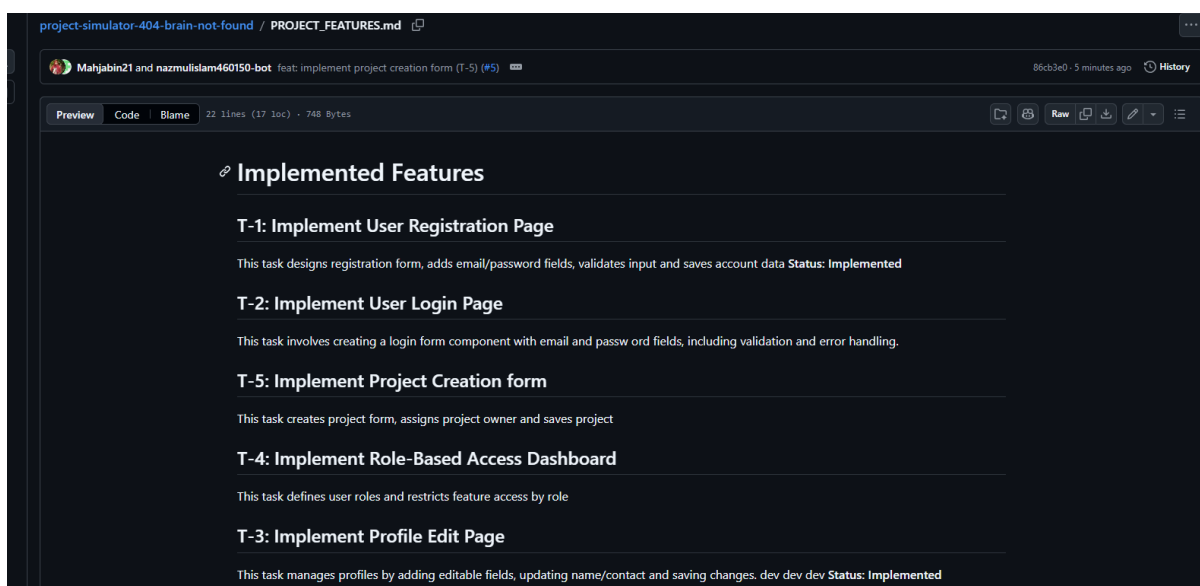


Fig: After merging the pull requests implemented features and their description

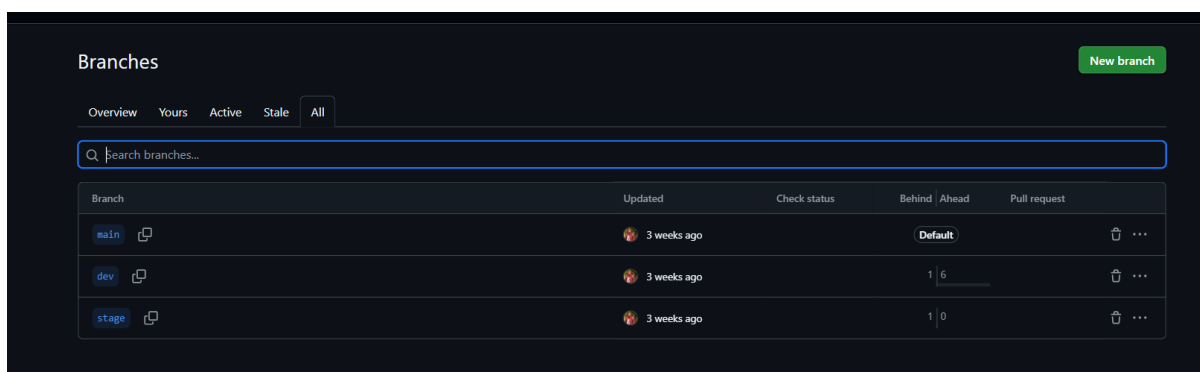


Fig: Deleted other branches

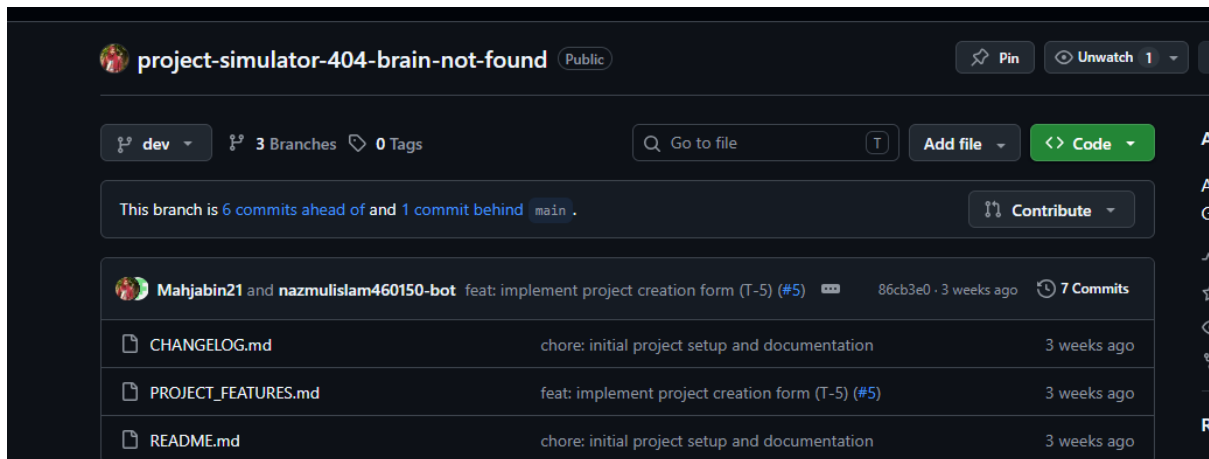


Fig: Updated features in the local dev branch

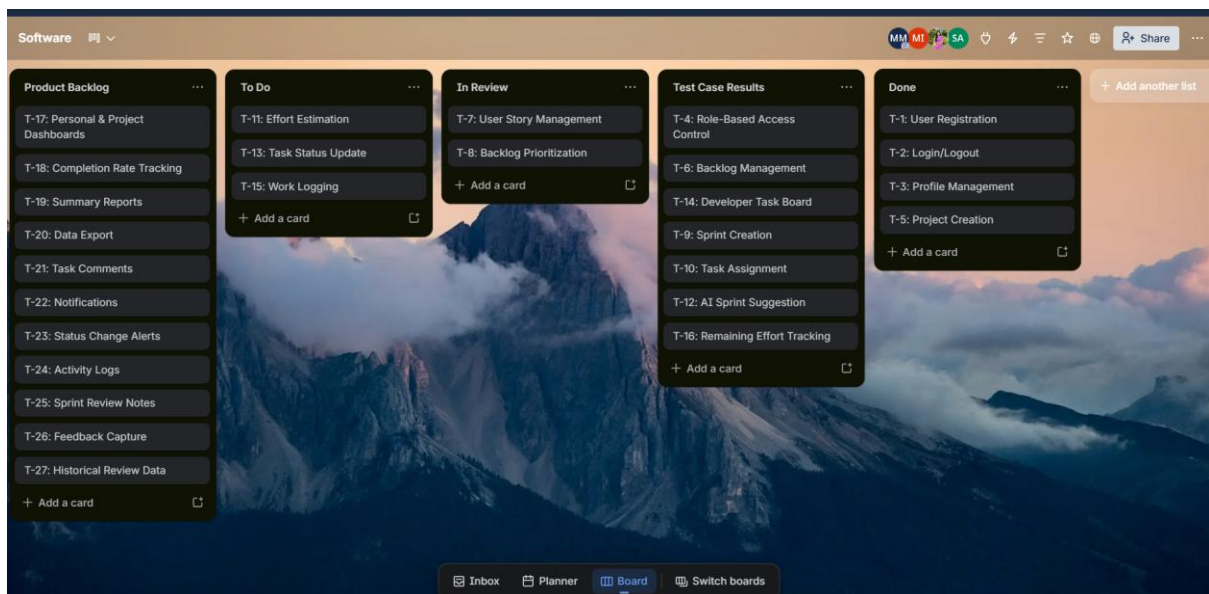


Fig: Features updates in Trello board

Github Link: <https://github.com/Mahjabin21/project-simulator-404-brain-not-found.git>

Trello Link:

<https://trello.com/invite/b/69d2a8016502b92047a719dd/ATTI5f66a9d02ad80ef72e9dbf748b01bfa598D73DAD/software>

6. SOFTWARE TESTING

Member 1: Maisha Mahjabin

Test 1:

Project Name: Development of a productivity management system for remote software teams			Test Designed by: Maisha Mahjabin	
Test Case ID: TC_Home_01			Test Designed date: 21/04/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Maisha Mahjabin	
Module Name: Homepage for all general users			Test Execution date: 22/04/2026	
Test Title: Verify homepage loads successfully				
Description: Test the website homepage				
Precondition: User has a stable internet connection				
Dependencies: if any				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open browser and enter website URL	URL	Homepage loads successfully	Homepage loaded without errors	Pass
2. Verify navigation bar (Home, Login, Sign Up)	-	All links visible and clickable	All links displayed correctly	Pass
3. Click Home	-	Reload homepage	Homepage reloaded correctly	Pass
4. Click Login	-	Redirect to login page	Login page opened successfully	Pass
5. Click Sign Up	-	Redirect to signup page	Signup page opened successfully	Pass
6. Click “Get Started” button	-	Redirect to signup/dashboard	Redirected to signup page	Pass
7. Check main heading and subheading	-	Text displayed correctly	Text displayed properly	Pass
8. Verify Key Features section	-	All feature cards visible	All cards displayed correctly	Pass

9. Click about link, privacy policy and contact	-	Open about, policy and contact page	About, policy and contact page opened correctly	Pass
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Test 2:

Project Name: Development of a productivity management system for remote software teams		Test Designed by: Maisha Mahjabin		
Test Case ID: TC_Login_02		Test Designed date: 21/04/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Maisha Mahjabin		
Module Name: Login Session		Test Execution date: 25/04/2026		
Test Title: Verify login with valid username and password				
Description: Test the website login page				
Precondition: The user has a valid username and password				
Dependencies: if any				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the site 2. Enter username 3. Enter password 4. Click submit	Username: Maisha Mahjabin Password: 1234 invisibly as ****	The user should log in to the application	Dashboard page opened successfully	Pass
5. Click Forgot Password	Email: maisha12@gmail.com Password: 54321 invisibly as *****	The user could reset the password if forgot it	Forgot Password page opened successfully	Pass

Member 2: Sadia Islam Anusha**Test 1:**

Project Name: Development of a productivity management system for remote software teams.			Test Designed by: Sadia Islam Anusha	
Test Case ID: TC_SprintBoard_03			Test Designed date: 22/04/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sadia Islam Anusha	
Module Name: Sprint Board			Test Execution date: 23/04/2026	
Test Title: Verify user can move tasks between columns on Sprint Board				
Description: Test the functionality of the Sprint Board to ensure users can view tasks and move them across columns (To Do, In Progress, Done)				
Precondition: The user has a valid username and password				
Dependencies: if any				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Sprint Board	—	Page layout is visible	Layout displayed correctly	Pass
2. Check side bar menu	—	Sidebar is aligned and visible	Sidebars correct	Pass
3. Enter task details	Assignee: Anusha	Task is created	Task created	Pass
4. Save task	-	Task appears in "To Do"	Task displayed	Pass
5. Click on a task	Task: "Fix Bug"	Task details open	Details displayed	Pass

Test 2:

Project Name: Development of a productivity management system for remote software teams			Test Designed by: Sadia Islam Anusha	
Test Case ID: TC_Admin_04			Test Designed date: 22/04/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Sadia Islam Anusha	
Module Name: Homepage for admin user			Test Execution date: 23/04/2026	
Test Title: Verify Admin Dashboard loads and core functionalities work correctly				
Description: Verify that the Admin Dashboard loads successfully, and all major components like user management, project controls, and reports are functional.				
Precondition: Admin user is logged into the system				
Dependencies: if any				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Login as Admin	Admin credentials	Admin dashboard loads successfully	Dashboard loaded without errors	Pass
2. Verify the sidebar menu (Dashboard, Projects, etc.)	-	All menu items visible and clickable	All menu items displayed correctly	Pass
3. Click "Projects"	-	Navigate to Projects page	Projects page opened successfully	Pass
4. Click "+ Create Project"	-	Project creation form opens	Form opened correctly	Pass
5. Click "Export Report"	-	Report is generated/downloaded	Report didn’t download successfully	Fail

6. Verify User Management table	-	User list displayed with Name, Email, Role, Status	User data displayed as UI implement	Pass
7. Click "Edit" on a user	-	Edit user form opens	Edit form didn't open successfully	Fail
8. Click "Delete" on a user	-	User is removed after confirmation	User didn't delete successfully	Fail
9. Check Recent Activity panel	-	Recent actions are displayed	Activity list didn't display correctly	Fail

Member 3: MD Nazmul Islam

Test 1:

Project Name: Development of a productivity management system for remote software teams			Test Designed by: MD Nazmul Islam	
Test Case ID: TC_Backlog_05			Test Designed date: 23/04/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: MD Nazmul Islam	
Module Name: Backlog Management			Test Execution date: 24/04/2026	
Test Title: backlog page loads and displays user stories				
Description: Test backlog page UI and user story list				
Precondition: User is logged in				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open backlog page	—	Backlog page loads	Loaded correctly	Pass
2. Check user story list	—	Stories displayed	Displayed correctly	Pass
3. Verify status labels	—	To Do/In Progress/Done visible	Not Correct	Fail
4.Click Add Story	—	Form opens	Opened	Pass
5.Submit new story	Sample story	Story added	Not Added	Fail

Test 2:

Project Name: Development of a productivity management system for remote software teams			Test Designed by: MD Nazmul Islam	
Test Case ID: TC_Project_06			Test Designed date: 23/04/2026	
Test Priority (Low, Medium, High): High			Test Executed by: MD Nazmul Islam	
Module Name: Project Dashboard			Test Execution date: 24/04/2026	
Test Title: Verify projects page loads and displays project list				
Description: Test the projects page UI and project list				
Precondition: User is logged in				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Open Projects page	—	Projects page loads successfully	Page loaded correctly	Pass
2. Check project list	—	Projects displayed	Displayed correctly	Pass
3. Verify project count card	—	Counts shown correctly	Not Correct	Fail
4. Enter project name	Website	Filtered results shown	Not work correctly	Fail
5. Clear search	—	All projects visible	Not Restored because UI element	Fail
6. Click Create Project	—	Form opens	Opened	Pass
7. Enter project details and submit	Sample Project	Project created	Created successfully	Pass
8. Select status filter	In Progress	Filtered projects shown	Correct	Pass
9. Change filter	Completed	Updated results	Not Correct	Fail

Member 4: Al Fardin Sowad

Test 1

Project Name: Development of a productivity management system for remote software teams			Test Designed by: Al Fardin Sowad	
Test Case ID: TC_UserDashboard_07			Test Designed date: 24/04/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Al Fardin Sowad	
Module Name: User Dashboard			Test Execution date: 25/04/2026	
Test Title: Verify dashboard loads and displays task summary				
Description: Test dashboard UI including task summary and task list				
Precondition: User is logged in				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Dashboard	—	Dashboard loads successfully	Loaded correctly	Pass
2. Check Total Tasks count	—	Correct number displayed	Correct	Pass
3. Check Completed Tasks	—	Correct number displayed	Correct	Pass
4. Check Pending Tasks	—	Correct number displayed	Correct	Pass
5. Verify task list and status	—	Tasks with status shown	Displayed correctly	Pass

Test 2:

Project Name: Development of a productivity management system for remote software teams			Test Designed by: Al Fardin Sowad	
Test Case ID: TC_Report_08			Test Designed date: 24/04/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Al Fardin Sowad	
Module Name: Reports			Test Execution date: 25/04/2026	
Test Title: Verify reports page and export functionality				
Description: Test reports listing and export feature				
Precondition: User is logged in				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Reports page	—	Reports page loads	Loaded correctly	Pass
2. Check recent reports list	—	Reports displayed	Reports displayed	Pass
3. Click Export Report	—	Report downloads as UI component	Didn’t download	Fail
4. Search report	Project Overview	Filtered result shown as UI component	Didn’t search	Fail

7. CONCLUSION

In conclusion, the development of the Productivity Management System for Remote Software Teams successfully addresses the key challenges faced in remote collaboration, such as task coordination, communication gaps, and progress tracking. By following the Scrum methodology, the project ensured a structured yet flexible development process, allowing continuous improvement through iterative sprints. The system integrates essential features like project and backlog management, sprint planning, task tracking, real-time dashboards, and role-based access control, which collectively enhance team efficiency and transparency. Additionally, the inclusion of AI-assisted functionalities, such as task recommendations and risk prediction, adds significant value by supporting smarter decision-making and better resource utilization. The testing results demonstrate that most core functionalities perform as expected, although a few areas require further refinement to achieve full reliability. Overall, this project meets its primary objectives by providing a centralized, user-friendly platform that improves productivity and collaboration among remote development teams. With further enhancements and optimization, the system has strong potential to be adapted for real-world applications and larger-scale software development environments.